ADVANCED MACHINE LEARNING · WORKSHEET 04

Multiple Linear Regression

Using the Ordinary Least Squares (OLS) Method

Lecture 4

Name	Branch / Year	Date
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SECTION 1 Starting with the Multiple Linear Regression Dataset

In Lecture 3, CGPA alone predicted the stipend. But what if IQ, attendance, and project scores also matter? One feature is not enough — we need many.

We write the data in this general form:

X_1	X_2	X_3	$\cdots$	X_m	Y	$\hat{Y}$
x_{11}	x_{12}	x_{13}	$\cdots$	x_{1m}	y_1	$\hat{y}_1$
x_{21}	x_{22}	x_{23}	$\cdots$	x_{2m}	y_2	$\hat{y}_2$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$
x_{n1}	x_{n2}	x_{n3}	$\cdots$	x_{nm}	y_n	$\hat{y}_n$

- m — number of features (input columns).
- n — number of data points (rows).
- Y — actual output values. $\hat{Y}$ — predicted output values.

The MLR prediction equation with m features:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_m x_m$$

★ KEY INSIGHT

This is the equation of a **hyperplane** in m dimensions. It is still a linear equation, only the number of input variables has increased from 1 to m.

PRACTICE P1 · MLR IN CONTEXT

A real-estate company wants to predict monthly apartment rent using four features: area (sq ft), number of bedrooms, distance to metro (km), and floor number.

(a) Write the MLR equation using meaningful symbols (x_{area}, x_{beds}, …). How many β parameters does the model have in total (including the intercept)?

ŷ = β0 + β1 x_area + β2 x_beds + β3 x_metro + β4 x_floor; 5 β parameters.

(b) Suppose you add a fifth feature (age of building in years). Must the value of β_1 (for area) stay the same as before? Why or why not?

No. Adding a feature refits all coefficients jointly, so correlations can change β1.

SECTION 2 Writing the Prediction Equation for Each Data Point

▷ HOOK

We have one equation template — but n data points. How do we write a prediction for every single row?

For the 1st data point:

$$\hat{y}_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \cdots + \beta_m x_{1m}$$

For the 2nd data point:

$\hat{y}_2 =$ ______ β0 + β1 x21 + β2 x22 + ... + βm x2m ______

For the i-th data point:

$\hat{y}_i =$ _β0_ $+$ _β1_ $x_{i1} +$ _β2_ $x_{i2} + \cdots +$ _βm_ x_{im}

Let us write all predicted values together as a vector:

$$\hat{\mathbf{y}} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \cdots + \beta_m x_{1m} \\ \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \cdots + \beta_m x_{2m} \\ \vdots \\ \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \cdots + \beta_m x_{nm} \end{bmatrix}.$$

This vector $\hat{\mathbf{y}}$ contains all predictions of the model.

★ KEY INSIGHT

Every predicted value has the same parameters $(\beta_0, \beta_1, \ldots, \beta_m)$, but the feature values change from row to row.

⧫ TAKEAWAY

The vector $\hat{\mathbf{y}}$ contains all n predictions of the model.

PRACTICE P2 · PREDICTIONS WITH NUMBERS

A teacher uses CGPA (x_1) and Study Hours per week (x_2) to predict exam scores. The trained model is $\hat{y} = 10 + 5\,x_1 + 2\,x_2$.

Student	CGPA (x_1)	Study Hrs (x_2)	Predicted Score ($\hat{y}$)
Arjun	8	6	62
Priya	7	10	65
Rahul	9	4	63

(a) Compute each student's predicted score using the model. (Fill the table above.)

(b) Who is predicted to score highest? Explain in one line.

Priya; her predicted score is 65, the highest.

(c) A new student Maya (CGPA = 8.5, Study Hrs = 8) joins. Can you predict her score with the same model, or must you retrain? Why?

Yes. ŷ = 10 + 5(8.5) + 2(8) = 68.5; a new prediction needs no retraining.

SECTION 3 Converting the Prediction Equations into Matrix Form

▷ HOOK

Writing n separate equations is tedious. Can we pack all of them into one single line using matrices?

Now let us convert this long system of equations into one compact matrix equation. The reason I do this is that matrix notation allows me to handle all n equations at once.

I can write

$$\hat{\mathbf{y}} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1m} \\ 1 & x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix}.$$

The first column of the matrix contains only 1's. I include this column because the intercept term β_0 must appear in every row.

Sizes:

- $\mathbf{X}$ (design matrix): $n \times (m+1)$ n rows, $m+1$ columns (1s column + m features).
- $\boldsymbol{\beta}$ (parameter vector): $(m+1) \times 1$.
- $\hat{\mathbf{y}}$ (prediction vector): $n \times 1$.

Compact form:

$$\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta}$$

★ KEY INSIGHT

Matrix notation compresses all n prediction equations into one line: $\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta}$.

⧫ TAKEAWAY

$\mathbf{X}$ is the **design matrix**, $\boldsymbol{\beta}$ is the **parameter vector**, $\hat{\mathbf{y}}$ is the **predicted vector**.

PRACTICE P3 · BUILDING THE DESIGN MATRIX

Using the same three-student dataset from P2 (model: $\hat{y} = 10 + 5\,x_1 + 2\,x_2$).

(a) Construct the design matrix $\mathbf{X}$ (remember the intercept column of 1s). State its dimensions.

$$\mathbf{X} = \begin{bmatrix} 1 & 8 & 6 \\ 1 & 7 & 10 \\ 1 & 9 & 4 \end{bmatrix}$$ Dimensions: 3×3

(b) Write $\boldsymbol{\beta}$ as a column vector and state its dimensions.

$$\boldsymbol{\beta} = \begin{bmatrix} 10 \\ 5 \\ 2 \end{bmatrix}$$ Dimensions: 3×1

(c) Verify: multiply the **first row** of $\mathbf{X}$ by $\boldsymbol{\beta}$. Do you get Arjun's predicted score from P2?

[1, 8, 6]β = 10 + 5(8) + 2(6) = 62. Yes.

(d) If a 4th student is added, what changes — the number of **rows** in $\mathbf{X}$, the number of **columns** in $\mathbf{X}$, or the length of $\boldsymbol{\beta}$?

Only X gains one row; its columns and the length of β stay unchanged.

SECTION 4 Defining the Residual at Each Data Point

▷ **HOOK**

Our model makes predictions, but how wrong are they? We need a way to measure the error at each point.

Residual (error) at the i-th data point:

$$e_i = y_i - \hat{y}_i$$

Squared error at the i-th data point:

$$e_i^2 = (y_i - \hat{y}_i)^2$$

Total squared error (the loss function to minimize):

$$L = e_1^2 + e_2^2 + \cdots + e_n^2 = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

★ **KEY INSIGHT**

Small residual = accurate prediction. Large residual = poor prediction. We denote this total squared error by $L(\boldsymbol{\beta})$ — this is the **loss function** we want to minimize.

PRACTICE P4 · COMPARING TWO MODELS

Two ML models predict monthly sales (in lakhs) for three stores:

Store	Actual	Model A	Model B	Residual (A / B)
Store 1	50	48	52	2 / -2
Store 2	30	36	28	-6 / 2
Store 3	45	44	46	1 / -1

(a) Compute the residual ($e = y - \hat{y}$) for each store under both models. (Fill the table above.)

(b) Model A's residual for Store 2 is negative (positive / negative).
This means Model A over-predicted (over-predicted / under-predicted) Store 2's sales.

(c) Compute the Total Squared Error (TSE) for each model:

$\text{TSE}_A = 2^2 + (-6)^2 + 1^2 = 41$

$\text{TSE}_B = (-2)^2 + 2^2 + (-1)^2 = 9$

Which model fits the data better? Model B

It has the smaller total squared error: 9 < 41.

SECTION 5 Writing the Residuals as a Vector

▷ **HOOK**

Instead of writing each residual separately, can we stack them into one vector and compute the total error in one shot?

Now let us write the error for each data point in vector form:

$$\mathbf{e} = \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ y_3 - \hat{y}_3 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix} = \mathbf{y} - \hat{\mathbf{y}}.$$

Now I multiply the residual vector by its transpose:

$$\mathbf{e}^\top \mathbf{e} = \big[\underline{\quad \text{y1 - ŷ1, y2 - ŷ2, ... , yn - ŷn} \quad}\big] \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}.$$

When I perform this multiplication, I get

$$\mathbf{e}^\top \mathbf{e} = (y_1 - \hat{y}_1)^2 + \underline{\quad \text{(y2 - ŷ2)^2} \quad} + \cdots + (y_n - \hat{y}_n)^2.$$

Therefore,

$$\mathbf{e}^\top \mathbf{e} = \sum_{i=1}^{n} (\underline{\quad \text{yi - ŷi} \quad})^2.$$

★ KEY INSIGHT

The dot product $\mathbf{e}^\top \mathbf{e}$ automatically computes the sum of squared residuals — one matrix operation replaces the entire summation.

This is the total squared error. Hence I can write the loss function as

$$L = \mathbf{e}^\top \mathbf{e}.$$

Now we substitute $\mathbf{e} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}$:

$$L(\boldsymbol{\beta}) = (\underline{\quad \text{y - Xβ} \quad})^\top (\underline{\quad \text{y - Xβ} \quad})$$

⧫ TAKEAWAY

The loss function $L(\boldsymbol{\beta}) = \mathbf{e}^\top \mathbf{e}$ takes the parameter vector $\boldsymbol{\beta}$ as input and returns one number (the total squared error) as output.

SECTION 6 Expanding the Loss Function

▷ HOOK

The loss $L(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ is compact but hard to differentiate. Let us expand it term by term.

Since $\mathbf{e} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}$:

$$L(\boldsymbol{\beta}) = (\underline{\quad \text{y - Xβ} \quad})^\top (\underline{\quad \text{y - Xβ} \quad})$$

Transpose of a difference: $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top = \underline{\quad \text{y^T} \quad} - \underline{\quad \text{β^T X^T} \quad}$

$$L(\boldsymbol{\beta}) = (\underline{\quad \text{y^T - β^T X^T} \quad})(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

$$= \underline{\quad \text{y^T y - y^T Xβ - β^T X^T y + β^T X^T Xβ} \quad}$$

Dimension verification:

- $\mathbf{y}^\top\mathbf{y} \to (1\times n)(n\times 1) = 1\times 1$ — scalar
- $\mathbf{y}^\top\mathbf{X}\boldsymbol{\beta}$ and $\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{y}$ — both scalars

Middle terms are equal scalars:

$$\mathbf{y}^\top\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{y} \qquad \text{(because the transpose of a scalar equals itself)}$$

Combining: $-\mathbf{y}^\top\mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{y} = -2\,\mathbf{y}^\top\mathbf{X}\boldsymbol{\beta}$

Expanded form (Equation 1):

$$\boxed{L(\boldsymbol{\beta}) = \mathbf{y}^\top\mathbf{y} - 2\,\mathbf{y}^\top\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta}}$$

This loss is a **convex function**. So when the loss is minimum, the partial derivatives with respect to all the parameters must become zero.

★ KEY INSIGHT

Three terms: $\mathbf{y}^\top\mathbf{y}$ (constant w.r.t. $\boldsymbol{\beta}$), $-2\,\mathbf{y}^\top\mathbf{X}\boldsymbol{\beta}$ (linear in $\boldsymbol{\beta}$), $\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta}$ (quadratic in $\boldsymbol{\beta}$). This loss is convex — the minimum exists and is unique.

⧫ TAKEAWAY

$L(\boldsymbol{\beta})$ maps a vector to a scalar: $L : \mathbb{R}^{m+1} \to \mathbb{R}$. At the minimum, all partial derivatives must be zero.

PRACTICE P5 · VERIFY THE EXPANSION WITH NUMBERS

Let $\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$, $\boldsymbol{\beta} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

(a) Compute $\mathbf{X}\boldsymbol{\beta}$ and then $\mathbf{e} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}$.

$\mathbf{X}\boldsymbol{\beta} =$ [3, 4]^T $\mathbf{e} =$ [2, 3]^T

(b) Compute $\mathbf{e}^\top\mathbf{e}$ (Total Squared Error) directly.

$\mathbf{e}^\top\mathbf{e} =$ 13

(c) Now compute each term of the expanded form separately:

$\mathbf{y}^\top\mathbf{y} =$ 74 $\mathbf{y}^\top\mathbf{X}\boldsymbol{\beta} =$ 43 $\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta} =$ 25

Verify: $\mathbf{y}^\top\mathbf{y} - 2(\mathbf{y}^\top\mathbf{X}\boldsymbol{\beta}) + \boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta} =$ 13.

Does it match part (b)? [Y]

(d) Confirm the two middle terms are equal: $\mathbf{y}^\top\mathbf{X}\boldsymbol{\beta} =$ 43, $\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{y} =$ 43.

Equal? [Y]

SECTION 7 Writing the Gradient Condition for the Minimum

▷ HOOK

At the bottom of a bowl, every direction is flat. In math terms: all partial derivatives equal zero.

At the minimum of $L(\boldsymbol{\beta})$:

$$\frac{\partial L}{\partial \beta_0} = 0, \quad \frac{\partial L}{\partial \beta_1} = 0, \quad \ldots, \quad \frac{\partial L}{\partial \beta_m} = 0$$

Stacked into the gradient vector:

$$\nabla L = \left[\frac{\partial L}{\partial \beta_0}, \frac{\partial L}{\partial \beta_1}, \ldots, \frac{\partial L}{\partial \beta_m}\right]^\top$$

At the minimum:

$$\frac{\partial L}{\partial \boldsymbol{\beta}} = \mathbf{0} \qquad \text{(zero vector)}$$

♦ TAKEAWAY

To differentiate Equation (1), we need three identities — one for each type of term (constant, linear, quadratic).

SECTION 8 Identity 1: Derivative of a Constant w.r.t. a Vector

▷ HOOK

If a function's output never changes no matter what $\boldsymbol{\beta}$ you plug in, what should its derivative be?

If $f(\boldsymbol{\beta}) = c$ (a constant scalar that does not depend on $\boldsymbol{\beta}$):

$$\frac{\partial f}{\partial \boldsymbol{\beta}} = \mathbf{0} \qquad \text{(zero vector)}$$

Applied to our loss: $\mathbf{y}^\top\mathbf{y}$ is independent of $\boldsymbol{\beta}$, so:

$$\frac{\partial(\mathbf{y}^\top\mathbf{y})}{\partial \boldsymbol{\beta}} = \mathbf{0} \qquad \cdots \text{ Equation (2)}$$

♦ TAKEAWAY

The derivative of any constant term with respect to $\boldsymbol{\beta}$ is the zero vector.

SECTION 9 Identity 2: Derivative of a Linear Form

▷ HOOK

$\mathbf{a}^\top\boldsymbol{\beta} = a_0\beta_0 + a_1\beta_1 + \cdots$ — differentiating each term gives back a_i. What does the full gradient look like?

Given $f(\boldsymbol{\beta}) = \mathbf{a}^\top\boldsymbol{\beta}$ where $\mathbf{a}$ is a constant vector:

$$f(\boldsymbol{\beta}) = a_0\beta_0 + a_1\beta_1 + \cdots + a_m\beta_m$$

Differentiating each component:

$$\frac{\partial f}{\partial \beta_0} = a_0, \quad \frac{\partial f}{\partial \beta_1} = a_1, \quad \ldots, \quad \frac{\partial f}{\partial \beta_m} = a_m$$

Stacked:

$$\frac{\partial(\mathbf{a}^\top\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \mathbf{a}$$

Applied to our loss: the linear term is $-2\,\mathbf{y}^\top\mathbf{X}\boldsymbol{\beta}$. Here $\mathbf{a}^\top = -2\,\mathbf{y}^\top\mathbf{X}$, so $\mathbf{a} = -2\,\mathbf{X}^\top\mathbf{y}$:

$$\frac{\partial(-2\,\mathbf{y}^\top\mathbf{X}\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -2\,\mathbf{X}^\top\mathbf{y} \qquad \cdots \text{ Equation (3)}$$

♦ TAKEAWAY

The derivative of the linear form $\mathbf{a}^\top\boldsymbol{\beta}$ with respect to $\boldsymbol{\beta}$ is simply the vector $\mathbf{a}$.

PRACTICE P6 · IDENTITIES 1 & 2 WITH NUMBERS

(a) Let $f(\boldsymbol{\beta}) = 42$ (a constant). What is $\partial f/\partial\boldsymbol{\beta}$? Why does this make sense intuitively?

0 (the zero vector); a constant does not change when β changes.

(b) Let $\mathbf{a} = [3,\ -1]^\top$. Write $f(\boldsymbol{\beta}) = \mathbf{a}^\top\boldsymbol{\beta}$ as a scalar expression: $f =$ 3β1 - β2.

Differentiate component-wise: $\partial f/\partial\beta_1 =$ 3, $\partial f/\partial\beta_2 =$ -1.

Stack them: $\partial f/\partial\boldsymbol{\beta} =$ [3, -1]^T. Does it equal $\mathbf{a}$? [Y]

(c) In our loss the linear term is $-2\,\mathbf{y}^\top\mathbf{X}\boldsymbol{\beta}$. If someone tells you that $\mathbf{X}^\top\mathbf{y} = [10,\ -4,\ 7]^\top$, what is the derivative of this term w.r.t. $\boldsymbol{\beta}$? (Don't forget the -2.)

-2 X^T y = [-20, 8, -14]^T

SECTION 10 Identity 3: Derivative of a Quadratic Form

▷ HOOK

$\boldsymbol{\beta}^\top\mathbf{A}\boldsymbol{\beta}$ is the matrix version of ax^2. The derivative of ax^2 is $2ax$. What is the matrix version?

Given $f(\boldsymbol{\beta}) = \boldsymbol{\beta}^\top\mathbf{A}\boldsymbol{\beta}$ where $\mathbf{A}$ is a constant symmetric matrix ($\mathbf{A}^\top = \mathbf{A}$).

Full 2×2 Worked Example

Let $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix}$ and $\boldsymbol{\beta} = [\beta_1,\ \beta_2]^\top$.

Step 1 — Compute $\mathbf{A}\boldsymbol{\beta}$:

$$\mathbf{A}\boldsymbol{\beta} = [a_{11}\beta_1 + a_{12}\beta_2,\quad a_{12}\beta_1 + a_{22}\beta_2]^\top$$

Step 2 — Compute $\boldsymbol{\beta}^\top(\mathbf{A}\boldsymbol{\beta})$:

$$\boldsymbol{\beta}^\top\mathbf{A}\boldsymbol{\beta} = a_{11}\beta_1^2 + 2a_{12}\beta_1\beta_2 + a_{22}\beta_2^2$$

Step 3 — Differentiate:

$$\frac{\partial f}{\partial\beta_1} = 2a_{11}\beta_1 + 2a_{12}\beta_2, \qquad \frac{\partial f}{\partial\beta_2} = 2a_{12}\beta_1 + 2a_{22}\beta_2$$

Step 4 — Stack and factor:

$$\frac{\partial f}{\partial\boldsymbol{\beta}} = 2\begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix}\begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} = \text{2Aβ}$$

General result:

$$\frac{\partial(\boldsymbol{\beta}^\top\mathbf{A}\boldsymbol{\beta})}{\partial\boldsymbol{\beta}} = \text{2Aβ}$$

Applying to Our Loss: The quadratic term is $\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta}$. Let $\mathbf{A} = \mathbf{X}^\top\mathbf{X}$.

Is $\mathbf{X}^\top\mathbf{X}$ symmetric? $(\mathbf{X}^\top\mathbf{X})^\top = \mathbf{X}^\top(\mathbf{X}^\top)^\top = \mathbf{X}^\top\mathbf{X} = \mathbf{A}$ ✓ Yes

Therefore:

$$\frac{\partial(\boldsymbol{\beta}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta})}{\partial\boldsymbol{\beta}} = 2\,\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta} \qquad \cdots \text{ Equation (4)}$$

♦ TAKEAWAY

The derivative of the quadratic form $\boldsymbol{\beta}^\top \mathbf{A}\boldsymbol{\beta}$ is $2\mathbf{A}\boldsymbol{\beta}$ — the matrix analogue of $d(ax^2)/dx = 2ax$.

PRACTICE P7 · IDENTITY 3 WITH A CONCRETE MATRIX

Let $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ and $\boldsymbol{\beta} = [\beta_1,\ \beta_2]^\top$.

(a) Verify that $\mathbf{A}$ is symmetric: does $\mathbf{A} = \mathbf{A}^\top$? [Y]

(b) Compute $\mathbf{A}\boldsymbol{\beta}$ (a 2×1 vector in terms of β_1, β_2):

$\mathbf{A}\boldsymbol{\beta} =$ [2β1 + β2, β1 + 3β2]^T

(c) Expand $f(\boldsymbol{\beta}) = \boldsymbol{\beta}^\top \mathbf{A}\boldsymbol{\beta}$ as a scalar expression in β_1, β_2:

$f =$ 2β1^2 + 2β1β2 + 3β2^2

(d) Differentiate component-wise:

$\partial f/\partial \beta_1 =$ 4β1 + 2β2 $\qquad$ $\partial f/\partial \beta_2 =$ 2β1 + 6β2

(e) Stack them as a vector and verify it equals $2\mathbf{A}\boldsymbol{\beta}$ from (b):

$\partial f/\partial \boldsymbol{\beta} =$ [4β1 + 2β2, 2β1 + 6β2]^T $= 2\mathbf{A}\boldsymbol{\beta}$? [Y]

SECTION 11 Differentiating the Expanded Loss Function

▷ HOOK

We have three identities and three terms. Time to assemble the pieces and find the minimum.

Differentiating Equation (1):

$$L(\boldsymbol{\beta}) = \underbrace{\text{y^T y}}_{\text{Eq 2}} - \underbrace{\text{2 y^T X}\beta}_{\text{Eq 3}} + \underbrace{\beta\text{^T X^T X}\beta}_{\text{Eq 4}}$$

Applying the three identities term by term:

$$\frac{\partial L}{\partial \boldsymbol{\beta}} = \text{0 - 2 X^T y + 2 X^T X}\beta = \text{-2 X^T y + 2 X^T X}\beta$$

At the minimum, $\partial L/\partial \boldsymbol{\beta} = \mathbf{0}$:

$$-\text{2 X^T y} + \text{2 X^T X}\beta = \mathbf{0}$$

Rearranging:

$$\boxed{\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta} = \mathbf{X}^\top \mathbf{y} \qquad \text{(Normal Equation)}}$$

Multiplying both sides by $(\mathbf{X}^\top \mathbf{X})^{-1}$:

$$\boxed{\boldsymbol{\beta}^* = (\text{X^T X})^{-1}\ \text{X^T y}}$$

★ KEY INSIGHT

This is the **closed-form OLS solution** for Multiple Linear Regression. It directly computes the optimal $\boldsymbol{\beta}$ from the data — no iteration, no guessing. The equation $\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta} = \mathbf{X}^\top \mathbf{y}$ is called the

Normal Equation.

♦ TAKEAWAY

$\boldsymbol{\beta}^* = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$ — the best parameter vector. Read it as: "compute $\mathbf{X}^\top\mathbf{X}$, invert it, multiply by $\mathbf{X}^\top\mathbf{y}$."

SECTION 12 When Does This Closed-Form Solution Work?

▷ HOOK

The formula looks clean. But does it always work?

For $\boldsymbol{\beta}^* = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$ to work, $(\mathbf{X}^\top\mathbf{X})^{-1}$ must exist. This requires the columns of $\mathbf{X}$ to be **linearly independent**. *(Note: The proof and significance of this condition will be covered in detail in upcoming lectures while discussing the Assumptions of Linear Regression.)* If $\det(\mathbf{X}^\top\mathbf{X}) = 0$, then $\mathbf{X}^\top\mathbf{X}$ is singular and the inverse does not exist.

△ WARNING MULTICOLLINEARITY

If any column of $\mathbf{X}$ is a linear combination of other columns, $\mathbf{X}^\top\mathbf{X}$ becomes singular. This is called **multicollinearity** — the closed-form solution breaks down. *This topic will be studied comprehensively in a later lecture.*

♦ TAKEAWAY

The OLS formula depends on the invertibility of $\mathbf{X}^\top\mathbf{X}$. Dependent columns → no inverse → no solution.

SECTION 13 Computational Limitation of the Closed-Form Solution

▷ HOOK

Even when the inverse exists, is it always practical to compute it?

$\mathbf{X}^\top\mathbf{X}$ has dimensions $(m+1)\times(m+1)$. Inverting a $k \times k$ matrix costs approximately:

$$O(k^3)$$

If $m = 10{,}000$ features:

$$10{,}000^3 = 10^{12} \text{ operations — one trillion!}$$

★ KEY INSIGHT

When the number of features is large, the closed-form solution becomes impractical. We then use iterative methods like **gradient descent** (next lecture).

♦ TAKEAWAY

Closed-form works for small m. For large m, use gradient descent.

SECTION 14 Proof of the $O(k^3)$ Inverse Complexity

▷ **HOOK**

Why exactly does inverting a $k \times k$ matrix take k^3 operations? Three nested loops tell the story.

To invert a $k \times k$ matrix $\mathbf{A}$, form the augmented matrix $[\mathbf{A} \mid \mathbf{I}]$ and row-reduce $\mathbf{A}$ into $\mathbf{I}$ (Gauss–Jordan elimination).

Three nested loops:

Loop	What It Does	Iterations
Outer	Iterate through k pivot columns	k
Middle	Eliminate k rows per pivot	k
Inner	Process $\sim 2k$ elements per row (augmented matrix)	$2k$

Total:

$$k \times k \times 2k = 2k^3 \to O(k^3)$$

♦ **TAKEAWAY**

Matrix inversion costs $O(k^3)$ due to three nested loops in Gauss–Jordan elimination.

PRACTICE P9 · REAL-WORLD SCENARIOS

(a) A startup collects three features for each data point: *temperature* (°C), *humidity* (%), and *temperature + humidity* (they literally summed the first two columns to create the third). Can they apply the OLS formula? If not, *what exactly goes wrong* and what is this problem called?

No. The third feature is the sum of the first two, so X^T X is singular

and cannot be inverted. This is perfect multicollinearity.

(b) Your company has a dataset with $m = 100{,}000$ features. Your computer performs 10^9 operations per second. Roughly how long would matrix inversion take? Is this practical?

Inversion cost $\approx$ (100,000)^3 = 10^15 operations.

Time $\approx$ 1,000,000 seconds $\approx$ 11.6 days.

Practical? No. What method would you use instead? Gradient Descent

(c) A colleague says: "Always use OLS — it gives the exact optimal answer!" Give **two** reasons why this is not always a good idea.

1. X^T X may be singular or numerically unstable.
2. Matrix inversion costs O(m^3), so it is impractical for many features.

SECTION 15 Final Takeaway — The Complete MLR-OLS Derivation

▷ **HOOK**

From a raw dataset to one elegant formula in 8 steps. Let us trace the path.

Step	Action	Result
1	Write MLR prediction	$\hat{y}_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_m x_{im}$
2	Stack as vector	$\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta}$
3	Define residual	$e_i = y_i - \hat{y}_i$
4	Total squared error	$L = \mathbf{e}^\top \mathbf{e} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$
5	Expand loss	$L = \mathbf{y}^\top \mathbf{y} - 2\,\mathbf{y}^\top \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}$
6	Derive 3 identities	Eq 2: $\partial(c)/\partial\boldsymbol{\beta} = \mathbf{0}$, Eq 3: $\partial(\mathbf{a}^\top\boldsymbol{\beta})/\partial\boldsymbol{\beta} = \mathbf{a}$, Eq 4: $\partial(\boldsymbol{\beta}^\top \mathbf{A}\boldsymbol{\beta})/\partial\boldsymbol{\beta} = 2\mathbf{A}\boldsymbol{\beta}$
7	Differentiate, set $\nabla = \mathbf{0}$	$\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta} = \mathbf{X}^\top \mathbf{y}$ (Normal Equation)
8	Solve for $\boldsymbol{\beta}^*$	$\boldsymbol{\beta}^* = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{y}$

TAKEAWAY

Same idea as SLR: minimize total squared error. With multiple features, matrices make it compact. The closed-form solution works when $\mathbf{X}^\top \mathbf{X}$ is invertible but costs $O(m^3)$.

Numerical Practice — OLS by Hand

Student	x_1	x_2	y
A	1	0	2
B	0	1	3
C	1	1	6

HOMEWORK P10 · FULL OLS COMPUTATION

STEP 1: Write $\mathbf{X}$ (with 1s column) and $\mathbf{y}$.

$$\mathbf{X} = \begin{bmatrix} 1 & \underline{1} & \underline{0} \\ 1 & \underline{0} & \underline{1} \\ 1 & \underline{1} & \underline{1} \end{bmatrix} \qquad \mathbf{y} = \begin{bmatrix} \underline{2} \\ \underline{3} \\ \underline{6} \end{bmatrix}$$

STEP 2: Compute $\mathbf{X}^\top$.

$$\mathbf{X}^\top = \begin{bmatrix} \underline{1} & \underline{1} & \underline{1} \\ \underline{1} & \underline{0} & \underline{1} \\ \underline{0} & \underline{1} & \underline{1} \end{bmatrix}$$

STEP 3: Compute $\mathbf{X}^\top \mathbf{X}$.

$(1,1) : 1+1+1 = \underline{3}$ $(1,2) : 1+0+1 = \underline{2}$ $(1,3) : 0+1+1 = \underline{2}$
$(2,1) : \underline{2}$ $(2,2) : 1+0+1 = \underline{2}$ $(2,3) : 0+0+1 = \underline{1}$
$(3,1) : \underline{2}$ $(3,2) : \underline{1}$ $(3,3) : 0+1+1 = \underline{2}$

$$\mathbf{X}^\top \mathbf{X} = \begin{bmatrix} \underline{3} & \underline{2} & \underline{2} \\ \underline{2} & \underline{2} & \underline{1} \\ \underline{2} & \underline{1} & \underline{2} \end{bmatrix}$$

Symmetric? $(1,2) = \underline{2}$ and $(2,1) = \underline{2}$. Equal? [Y]

STEP 4: Compute $\mathbf{X}^\top \mathbf{y}$.

Row 1: $1(2) + 1(3) + 1(6) = \underline{11}$

Row 2: $1(2) + 0(3) + 1(6) =$ __8__
Row 3: $0(2) + 1(3) + 1(6) =$ __9__
$\mathbf{X}^\top \mathbf{y} = [$ __11__, __8__, __9__ $]^\top$

STEP 5: $\det(\mathbf{X}^\top\mathbf{X}) =$ __3__(__4__ − __1__) − __2__(__4__ − __2__) + __2__(__2__ − __4__) = __1__

Invertible? [Y]

STEP 6:

$$(\mathbf{X}^\top\mathbf{X})^{-1} = \begin{bmatrix} 3 & -2 & -2 \\ -2 & 2 & 1 \\ -2 & 1 & 2 \end{bmatrix}$$

STEP 7: $\boldsymbol{\beta}^* = (\mathbf{X}^\top\mathbf{X})^{-1} \times \mathbf{X}^\top\mathbf{y}$

$\beta_0 =$ __-1__ $\quad \beta_1 =$ __3__ $\quad \beta_2 =$ __4__

STEP 8: Final model: $\hat{y} =$ __-1__ + __3__ x_1 + __4__ x_2

STEP 9: Verify.

A: __-1__ + __3__(1) + __4__(0) = __2__ $\quad (y = 2)$ [Y]
B: __-1__ + __3__(0) + __4__(1) = __3__ $\quad (y = 3)$ [Y]
C: __-1__ + __3__(1) + __4__(1) = __6__ $\quad (y = 6)$ [Y]

Answers at the end.

Reflect 1 We had 3 data points and 3 parameters — perfect fit. With 100 data points but 3 parameters, would it still perfectly fit?
Hint: n equations vs $(m+1)$ unknowns. Perfect fit requires $n \le (m+1)$.

Reflect 2 The formula gives the exact answer in one step. Gradient Descent takes many steps. Why might we prefer the slow method?
Hint: What if $m = 10{,}000$?

SLR vs MLR Comparison

Aspect	SLR (Lecture 3)	MLR (Lecture 4)
Notation	$m,\ c$	$\beta_0,\ \beta_1,\ \ldots,\ \beta_m$
Model	$\hat{y} = mx + c$	$\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta}$
Error	$\sum(y_i - mx_i - c)^2$	$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$
Minimize	$\partial L/\partial m = 0,\ \partial L/\partial c = 0$	$\partial L/\partial \boldsymbol{\beta} = \mathbf{0}$
Solution	$m,\ c$ formulas	$\boldsymbol{\beta}^* = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$
Geometry	Line	Hyperplane
Condition	$\sum(x_i - \bar{x})^2 \neq 0$	$\det(\mathbf{X}^\top\mathbf{X}) \neq 0$
Limitation	1 feature	$O(m^3)$ cost

HOMEWORK P11 · TRUE OR FALSE — JUSTIFY THE TRICKY ONES

For each statement write **T** or **F**. Where it says "Justify", add a one-line explanation.

(a) SLR is a special case of MLR with $m = 1$. [T]

(b) If you double every value in column x_2 of $\mathbf{X}$, the OLS formula still produces a valid $\boldsymbol{\beta}^*$ (assuming no other issues). [T]
Justify: Doubling x2 halves its coefficient; the fitted predictions remain unchanged.

(c) OLS minimizes the sum of *absolute* residuals. [**F**]

(d) If $\mathbf{X}$ has 200 rows and 5 features, then $\mathbf{X}^\top\mathbf{X}$ is a 6×6 matrix. [**T**]
Justify: X has 6 columns including the intercept, so X^T X is 6 × 6.

(e) Multicollinearity means two features are *slightly correlated.* [**F**]

(f) If Gradient Descent and OLS are both applicable, they will find *different* optimal $\boldsymbol{\beta}^*$ values. [**F**]
Justify: For the same convex loss, converged GD and OLS reach the same optimum.

(g) Adding more features to a model *always* reduces the Total Squared Error on the training data. [**F**]
Justify: Training TSE cannot increase, but it may stay unchanged rather than decrease.

(h) The Normal Equation is $\mathbf{X}^\top\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^\top\mathbf{y}$. [**T**]

(i) If $n < (m+1)$, the model can perfectly fit the training data, and this is always a good thing. [**F**]
Justify: It may interpolate the data, but can overfit and generalize poorly.

BY THE END OF THIS SHEET, YOU CAN

- ☐ Extend SLR to MLR and construct the design matrix $\mathbf{X}$ with the intercept column of 1s.
- ☐ Express the loss as $L(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ and expand it into three terms.
- ☐ Derive three matrix calculus identities (constant, linear form, quadratic form).
- ☐ Apply identities to derive the closed-form OLS solution: $\boldsymbol{\beta}^* = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$.
- ☐ Explain when the solution fails (multicollinearity) and why inversion costs $O(m^3)$.
- ☐ Compute $\boldsymbol{\beta}^*$ by hand for a small dataset using matrix operations.